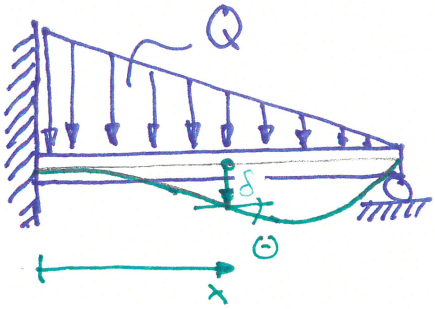
Elementarfall

Lösta fall (mha elast. linj. diff.ekv.):
Balkar (L, EI) med olika laster och RV,
map deformationer, stödreaktioner mm.

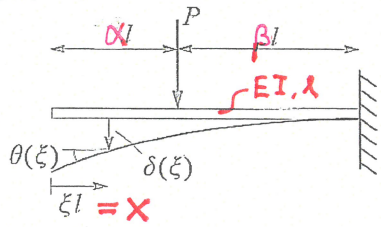
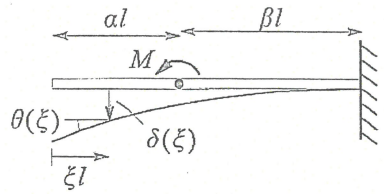
I handböcker, tabeller, formelsamlingen:

Ex:



356 KAP 32 ELEMENTARFALL →357

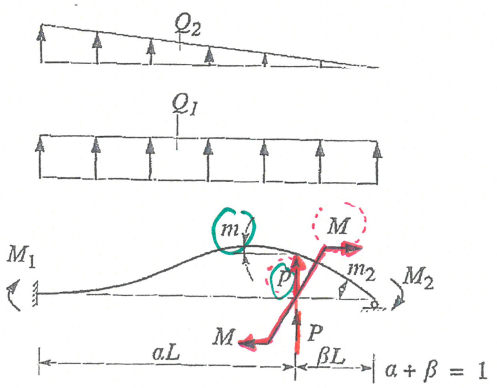
bell 32.1 Elementarfall för konsolbalk

Belastningsfall, $\alpha+\beta=1$	θ = vinkeländring, $\delta(\xi=x/l)$ = förskjutning
Lastens position  Var hänstams?	$\theta(\xi<\alpha)=\frac{Pl^2}{2EI}\beta^2$ $\delta(\xi<\alpha)=\frac{Pl^3}{6EI}(-\beta^3+3\beta^2(1-\xi))$ ← till vänster om P $\theta(\xi>\alpha)=\frac{Pl^2}{2EI}(\beta^2-(\xi-\alpha)^2)$ $\delta(\xi>\alpha)=\frac{Pl^3}{6EI}((\xi-\alpha)^3-3\beta^2(\xi-\alpha)+2\beta^3)$ ← till höger om P $\theta(\alpha)=\frac{Pl^2}{2EI}\beta^2 \quad \delta(\alpha)=\frac{Pl^3}{3EI}\beta^3$
 	$\theta(\xi<\alpha)=\frac{Ml}{EI}\beta$ $\delta(\xi<\alpha)=\frac{Ml^2}{EI}\beta(1-\xi-\beta/2)$ $\theta(\xi>\alpha)=\frac{Ml}{EI}(1-\xi)$ $\delta(\xi>\alpha)=\frac{Ml^2}{2EI}((\xi-\alpha)^2-2\beta(\xi-\alpha)+\beta^2)$ $\theta(\alpha)=\frac{Ml}{EI}\beta \quad \delta(\alpha)=\frac{Ml^2}{2EI}\beta^2$

FS 358-359

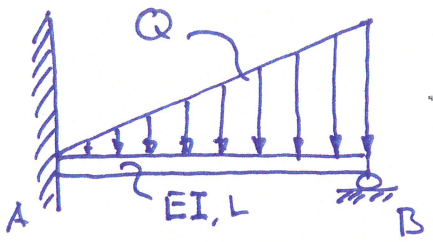
Tex: P = M [ ] proportionalitetskonstant(er) mellan p och M (mfl laster)

Fast inspänd - glidledad balk.



	M_2	M	P	Q_1	Q_2
M_1	$\frac{1}{2}$	$\frac{1}{2}(1-3\beta^2)$	$\frac{L}{2}\alpha\beta(1+\beta)$	$\frac{L}{8}$	$\frac{2L}{15}$
m_2	$\frac{L}{4EI}$	$\frac{L}{4EI}\alpha(1-3\beta)$	$\frac{L^2}{4EI}\alpha^2\beta$	$\frac{L^2}{48EI}$	$\frac{L^2}{60EI}$
m	$\frac{L}{4EI}\alpha(1-3\beta)$	$\frac{L}{4EI}\alpha(\alpha^3+4\beta^3)$	$\frac{L^2}{4EI}\alpha^2\beta(\alpha^2-2\beta^2)$	–	–
p	$\frac{L^2}{4EI}\alpha^2\beta$	$\frac{L^2}{4EI}\alpha^2\beta(\alpha^2-2\beta^2)$	$\frac{L^3}{12EI}\alpha^3\beta^2(3+\beta)$	–	–

Elementarfall kan sättas ihop och superponeras för att lösa fler (och mer komplic.) fall. Exempel:

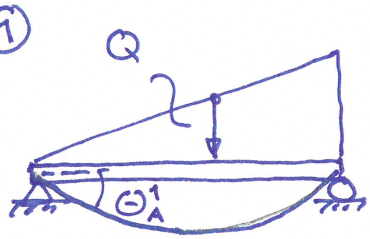


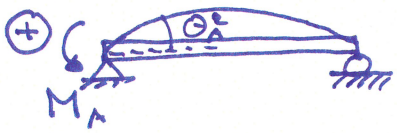
Sök (tex) inspänningsmomentet M_A

Finns ej som fall i FS.

Men kan konstrueras enl:

$\equiv$

① 

② 

Krav: Fast insp. vid A

dvs $\Theta_A^1 = \Theta_A^2$ eller:

$$\frac{7QL^2}{180EI} = \frac{M_A L}{3EI}$$

varav $M_A = \frac{7QL}{60}$

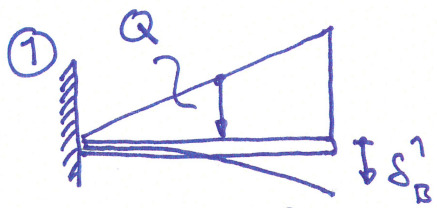
FS s ~~345~~ 357 fall 4

$$\Theta_A^1 = \frac{7QL^2}{180EI}$$

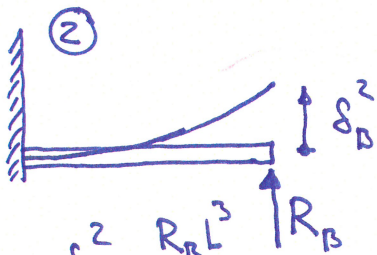
fall 5 $= 0$

$$\Theta_A^2 = \frac{M_A L}{3EI} + \frac{M_B L}{6EI}$$

Alternativt $\equiv$

① 

$$\delta_B^1 = \frac{11QL^3}{60EI}$$

② 

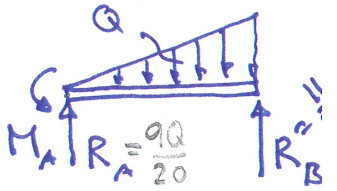
$$\delta_B^2 = \frac{R_B L^3}{3EI}$$

Stöd vid B (ingen utböj.) kräver $\delta_B^1 = \delta_B^2$ dvs

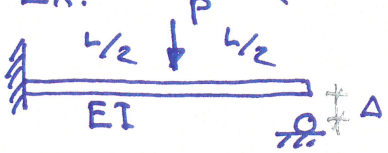
$$\frac{11QL^3}{60EI} = \frac{R_B L^3}{3EI}$$

varav $R_B = \frac{11Q}{20}$

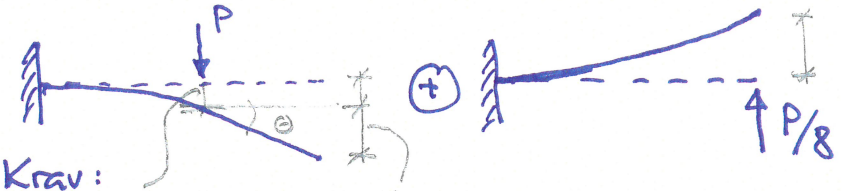
Momentjämvikt ger nu $M_A = Q \cdot \frac{2L}{3} - \frac{11Q}{6} \cdot L$: $\frac{40-33}{60}$ av



Ex. (tentatal)



$\equiv$



Vilket P ger stödreaktion $R_B = \frac{P}{8}$ (efter kontakt)?

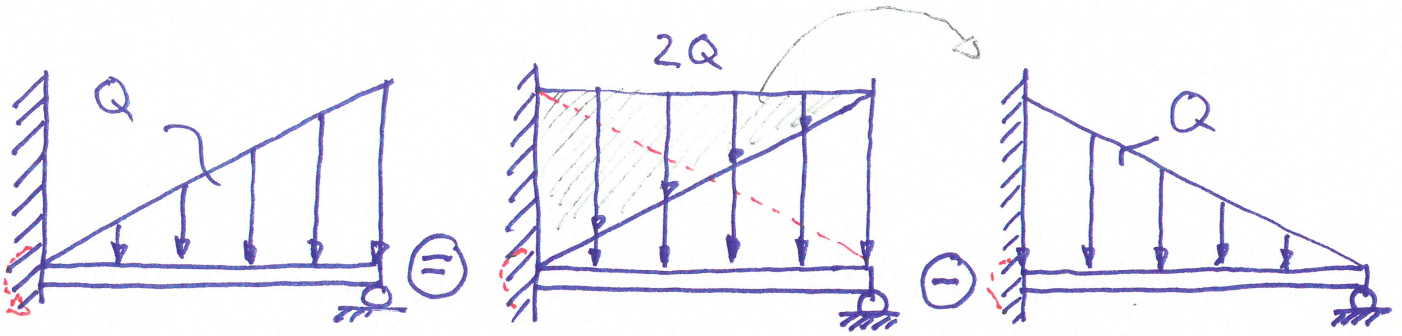
Krav:

$$\Delta = \frac{P(L/2)^3}{3EI} + \frac{P(L/2)^2}{2EI} \cdot \frac{L}{2} - \frac{(P/8)L^3}{3EI} = \frac{PL^3}{16EI}$$

varav $P = \frac{16EI}{L^3}\Delta$ (även FS 356 fall 9, $\alpha = \beta = \frac{1}{2}$, $\xi = 0$)

Alt. 3

Ta bort: Motsv.

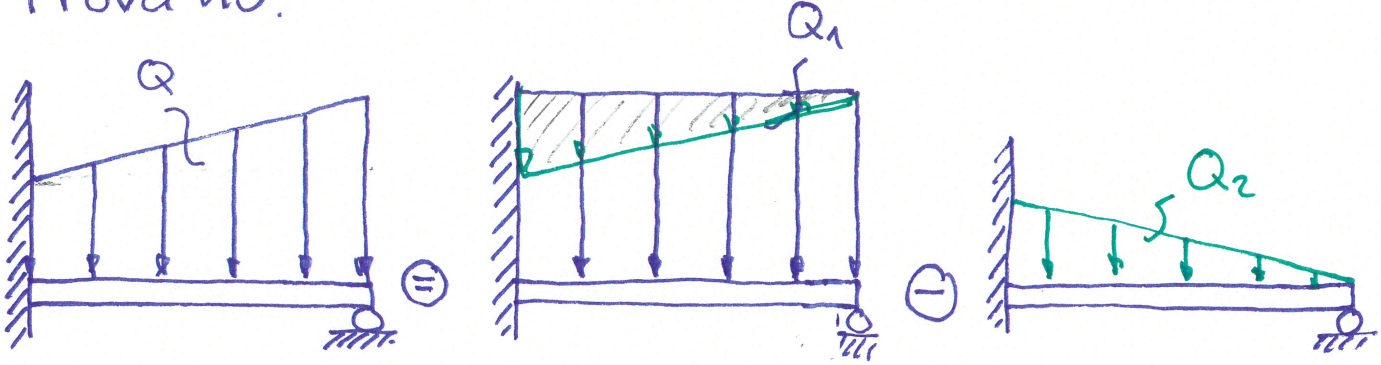


$M_A = ?$ $= \frac{2QL}{8} - \frac{2QL}{15} = \frac{7QL}{60}$

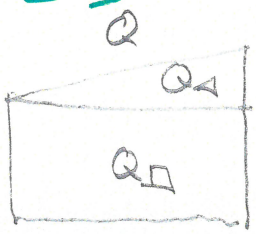
$\frac{15QL}{60}$ $\quad - \frac{8QL}{60}$

SSB

Prova nu:

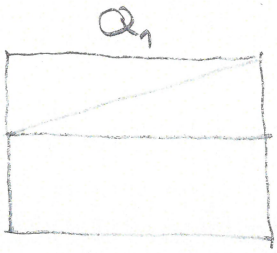


Godtyckl. rät-linjig utbr. last:



$Q = Q_{\square} + Q_{\triangle}$

Hur bestämma Q_1 och Q_2?



$Q_1 = Q_{\square} + 2Q_{\triangle}$

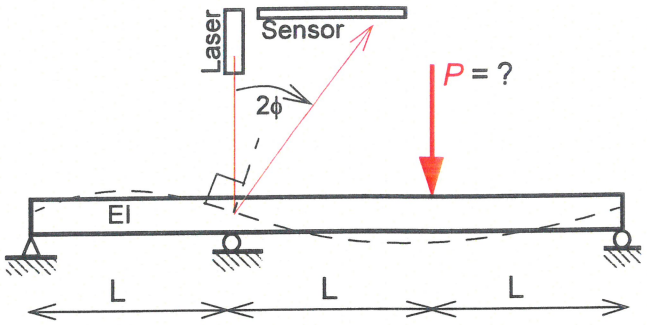


$Q_2 = Q_{\triangle}$

3. En balk, böjstyvhet EI, fritt upplagd på tre stöd belastas med en kraft P mitt i högra spannet enligt figuren. Med hjälp av en reflekterad laserstråle kan små vinkeländringar mätas mycket noggrant.

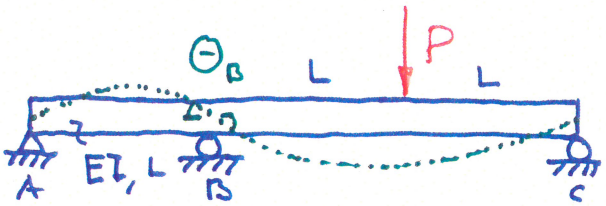
Bestäm lasten P om sensorn mäter den reflekterade strålens vinkel till 2ϕ. (6p)

Tentamen 210911

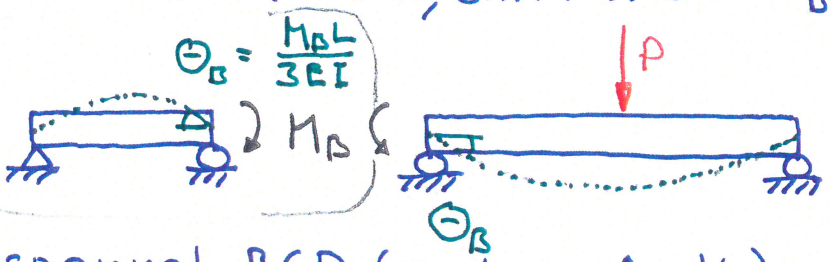


3) Sökt är Θ_B, där spann AB utgör en torsionsfjäder med styvhet:

$$k_\Theta = \frac{3EI}{L}$$



Snitta vid B, snittmoment M_B



Teckna Θ_B för spannet BCD (med $\alpha = \beta = 1/2$)

$$\Theta_B = -\frac{M_B \cdot 2L}{3EI} + \frac{P(2L)^2}{6EI}\alpha\beta(1+\beta) \quad (3/8) = \{M_B = k_\Theta \cdot \Theta_B\}$$

OBS

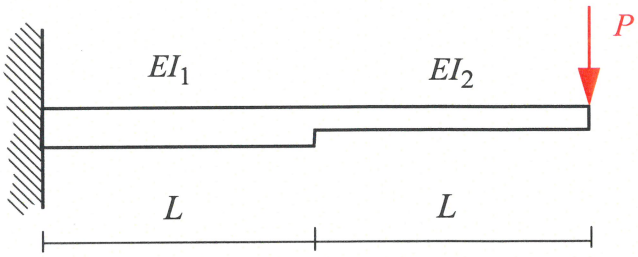
$$= -\frac{2L}{3EI}\cdot\frac{3EI}{L}\Theta_B + \frac{PL^2}{4EI} \Leftrightarrow 3\Theta_B = \frac{PL^2}{4EI}$$

Läs ut

$$\boxed{P = \frac{12EI}{L^2}\Theta_B = \frac{12EI}{L^2}\varphi}$$

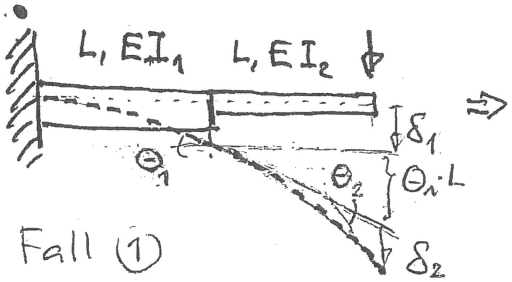
Halva reflektionsvinkeln

3. En konstruktion består av två balkar med olika böjstyvheter (EI) och är fast inspänd till vänster, se figuren. Beräkna nedböjning δ och rotation θ längst ut där kraften P angriper.

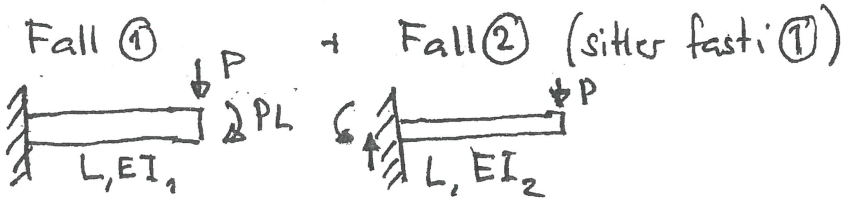


Balktal: Bestäm δ och Θ vid P

3.



$\Rightarrow$



Fall ①

$$\delta_1 = \frac{PL^3}{3EI_1} + \frac{PL\cdot L^2}{2EI_1} = \frac{5}{6}\frac{PL^3}{EI_1}$$

$$\Theta_1 = \frac{PL^2}{2EI_1} + \frac{PL\cdot L}{EI_1} = \frac{3}{2}\frac{PL^2}{EI}$$

Fall ②

$$\delta_2 = \frac{PL^3}{3EI_2}$$

$$\Theta_2 = \frac{PL^2}{2EI_2}$$

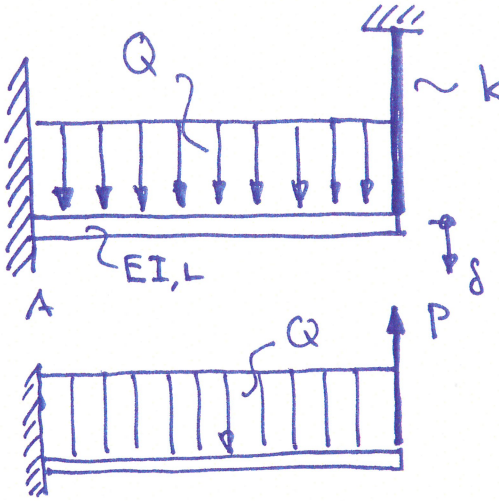
Totalt

$$\delta_1 + \underbrace{\Theta_1 L}_{\text{OBS}} + \delta_2 = \frac{PL^3}{6E}\left(\frac{14}{I_1} + \frac{2}{I_2}\right)$$

$$\Theta_1 + \Theta_2 = \frac{PL^2}{2E}\left(\frac{3}{I_1} + \frac{1}{I_2}\right)$$

Exempel:

Konsolbalk med lina / fjädrande stöd



$\sim k = \frac{(EA)_e}{L_e}$

Hur beror δ och M_A av linans/stödets styvhet?

Elementarfall (1 & 3 s344) ger:

$$\delta = \frac{QL^3}{8EI} - \frac{PL^3}{3EI}$$ där linkraften är $P = k\cdot\delta$

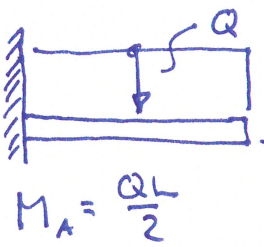
Dvs $\delta = \frac{QL^3}{8EI} - k\cdot\delta\frac{L^3}{3EI} \iff \delta\left(1+\frac{kL^3}{3EI}\right) = \frac{QL^3}{8EI}$

varav $\delta = \frac{3Q}{8\left(\frac{3EI}{L^3}+k\right)} \to \frac{QL^3}{8EI}$ då $k\to 0$ (och $\to 0$ då $k\to\infty$)

Moment: $M_A = \frac{QL}{2} - PL = \frac{QL}{2}\left[1 - \frac{6}{8\left(1+\frac{3EI}{kL^3}\right)}\right] \to \frac{QL}{2}$ $k\to 0$; $\to \frac{QL}{8}$ $k\to\infty$

($PL = k\cdot\delta$)

Fallen kan kontrolleras, finns i FS s 346-7

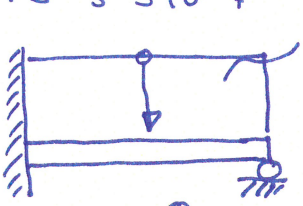


$k = 0$

$\delta = \frac{QL^3}{8EI}$

$M_A = \frac{QL}{2}$

resp

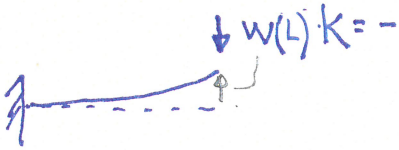


$k\to\infty$

$M_A = \frac{QL}{8}$

Kan även lösas mha elastiska linjen: $EIw^{IV}(x) = -\frac{Q}{L}$

$$EIw(x) = -\frac{Qx^4}{24EI} + Ax^3 + Bx^2 + Cx + D$$

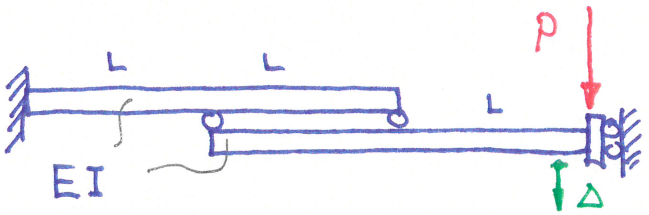


med RV:

$x=0$ (fast insp.): $\begin{cases} w = 0 \\ w' = 0 \end{cases}$

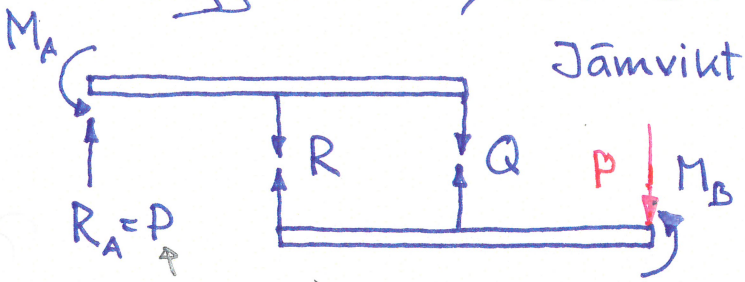
$x=L$: $\begin{cases} w'' = 0 \text{ ty } M = 0 \\ T(L) = -EIw'''(L) = -k\cdot w \end{cases}$ ← fjädrande stöd / elast. lina

Exempel (tentatal): Bestäm styvheten P/Δ.



Notera likformigheten i last o deformation för de två balkarna.

Frilägg delarna, inför snitt- och reaktionslaster:



Jämvikt - Övre: $\uparrow$ $P - R - Q = 0$ ①

$\circlearrowleft_A$ $M_A - RL - Q \cdot 2L = 0$ ②

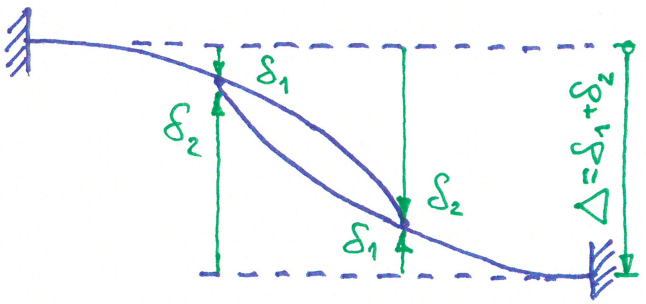
Nedre: $\uparrow$ $R + Q - P = 0$ ③ = ①

$\circlearrowleft_B$ $M_B - Q \cdot L - R \cdot 2L = 0$ ④

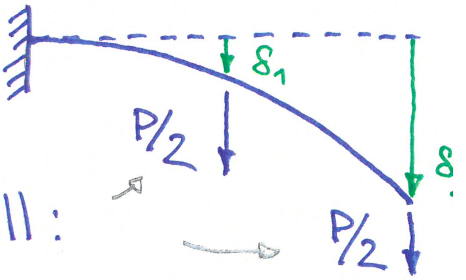
(4 obek, bara 3 ekv. Deformationsvillkor behövs)

Ansätt Δ (Likformighet: Balkdeform. lika, ger $R = Q = \frac{P}{2}$)

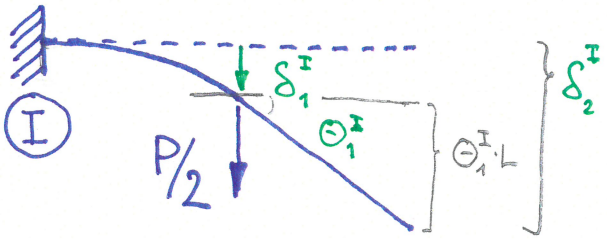
och: $M_A = M_B = 3PL/2$

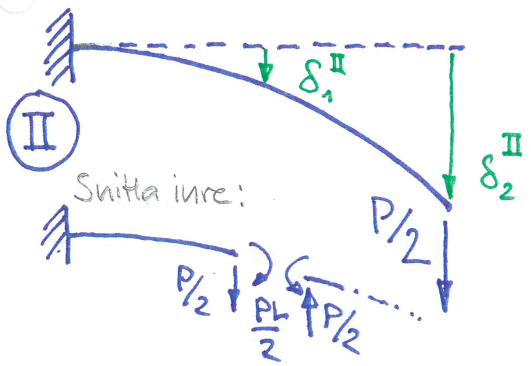


Betrakta (tex) övre



Två lastfall:





Kräv: 4 termer (elementarfall)

$$\delta_1^I = \frac{(P/2)L^3}{3EI} = \frac{PL^3}{6EI}$$

$$\delta_2^I = \delta_1^I + \Theta_1^I \cdot L = \frac{PL^3}{6EI} + \frac{(P/2)L^2}{2EI} L = \frac{5PL^3}{12EI}$$

$$\delta_2^{II} = \frac{(P/2)(2L)^3}{3EI} = \frac{4PL^3}{3EI}$$

inre: $\delta_1^{II} = \frac{(P/2)L^3}{3EI} + \frac{(PL/2)L^2}{2EI} = \frac{5PL^3}{12EI}$

pga. kraft, moment

Totalt: $\Delta = \delta_1 + \delta_2 = \delta_1^I + \delta_1^{II} + \delta_2^I - \delta_2^{II} = \frac{PL^3}{12EI}(2 + 5 + 5 + 16) = \ldots = \frac{7PL^3}{3EI}$

Varav: $\frac{P}{\Delta} = \frac{3EI}{7L^3}$